

Module Title: VLANs

Module Objective: Explain the purpose and operation of VLANs in a switched network and their configuration

Topic Title	Topic Objective
Overview of VLANs	Explain the purpose of VLANs in a switched network.
VLANs in a Multi-Switched Environment	Explain how a switch forwards frames based on VLAN configuration in a multi-switch environment.
VLAN Configuration	Configure a switch port to be assigned to a VLAN based on requirements.
VLAN Trunks	Configure a trunk port on a LAN switch.
Dynamic Trunking Protocol	Configure Dynamic Trunking Protocol (DTP).

Page #	Activity Type	Activity Name
3.1.4	Packet Tracer	Who Hears the Broadcast?
3.1.5	Check Your Understanding	Overview of VLANs

3.2.8	Packet Tracer	Investigate a VLAN Implementation
3.2.9	Check Your Understanding	VLANs in a Multi-Switched Environment
3.3.11	Syntax Checker	VLAN Configuration
3.3.12	Packet Tracer	VLAN Configuration
3.4.5	Packet Tracer	Configure Trunks
3.4.6	Lab	Configure VLANs and Trunking
3.5.5	Packet Tracer	Configure DTP
3.5.6	Check Your Understanding	Dynamic Trunking Protocol
3.6.1	Packet Tracer	Implement VLANs and Trunking
3.6.2	Lab	Implement VLANs and Trunking

New Terms

VLAN

Logical broadcast domain

Data VLAN

Default VLAN

Native VLAN

Management VLAN

show vlan brief

Voice VLAN

VLAN Trunk

VLAN Segmentation

IEEE 802.1Q

VLAN Tagging

Canonical Format Identifier (CFI)

User Priority

VLAN ID

Type

show interfaces int switchport

Normal Range VLANs

Extended Range VLANs

vlan vlan-id

name vlan-name

switchport mode access

switchport access vlan vlan-id

interface range

no switchport access vlan vlan-id

no vlan vlan-id

delete flash:vlan.dat

delete vlan.dat

show vlan

show interfaces

show vlan summary

show interfaces vlan vlan_id

switchport mode trunk

switchport trunk allowed vlan vlan_list

switchport trunk native vlan vlan_id

no switchport trunk allowed vlan

no switchport trunk native vlan

show interfaces switchport

no switchport access vlan vlan_id

show interfaces trunk

show interfaces int_id trunk